

- 5 (a) field into (the plane of) the paper B1 [1]
- (b) force due to magnetic field provides the centripetal force B1
 $mv^2 / r = Bqv$ C1
 $B = (20 \times 1.66 \times 10^{-27} \times 1.40 \times 10^5) / (1.6 \times 10^{-19} \times 6.4 \times 10^{-2})$ B1
 $= 0.454 \text{ T}$ A0 [3]
- (c) (i) semicircle with diameter greater than 12.8 cm B1 [1]
- (ii) new flux density = $\frac{22}{20} \times 0.454$ C1
 $B = 0.499 \text{ T}$ A1 [2]
- 6 (a) (i) e.g. prevent flux losses / improve flux linkage B1 [1]